CTF-GS: Compact Teacher Feature Fields with View-to-Primitive Registration for Open-Vocabulary 3D Gaussian Scene Understanding

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Abstract

We study whether dense vision-foundation features can be reconstructed as a 3D Gaussian scene representation and reused at novel views for open-vocabulary scene understanding. Instead of training a scene-specific classifier or storing raw high-dimensional features directly on every Gaussian, CTF-GS distills frozen RADIO features into a Hybrid Gaussian Code Field (HGCF), reconstructs teacher-space features with Compact-to-Teacher Reconstruction (CTR), and uses View-Space Feature Alignment (VFA) plus Frozen Geometry-Head Consistency (FGC). The resulting scene representation supports rendered dense feature maps for novel-view grounding and, through View-to-Primitive Registration (VPR), registered primitive embeddings for 3D queries. On LERF-OVS, the current frozen RADIO-GS package reaches 0.8712 macro localization accuracy and 0.5243 calibrated macro mIoU across four scenes. On a 10-scene ScanNet direct point-query protocol, a conservative Gaussian-index readout reaches 0.3538, 0.3573, and 0.4293 mIoU on the 19-, 15-, and 10-class splits; a contextual kNN point readout with label-free scene-mean calibration further improves them to 0.3637, 0.3708, and 0.4512. An OpenGaussian-style direct 3D object-selection evaluation shows that a rendered-feature-to-primitive registration readout with GT-free voxel context aggregation and a fixed mean+2.5std score-distribution selector with a 0.5% floor and 1.8% top-ratio cap reaches 0.4227 macro mIoU and 0.6906 macro Acc@0.25; the optional GT-free RGB boundary snap used for rendered-mask projection improves this to 0.4554 macro mIoU and 0.7014 Acc@0.25. For context, the raw Gaussian-center diagnostic reaches only 0.012/0.009 under the fixed top-2% selector used in the VPR sweep and 0.0804/0.0932 under an earlier top-10% selector. These results support the central claim that 3D Gaussian scenes can serve as reusable foundation-feature memories across rendered-view and registered primitive-level readouts, while clarifying that raw Gaussian-center readout is weak and Waldo Kitchen remains a difficult object-fragmentation case.

1 Introduction

Vision foundation models expose dense features that are useful for recognition, localization, segmentation, and geometric reasoning. However, most 3D scene representations still optimize either RGB appearance or task-specific labels. In open-vocabulary 3D understanding, this creates a gap: the strongest 2D feature extractors are available at training images, but a deployed 3D scene must answer queries from novel views.

CTF-GS targets this gap by reconstructing teacher foundation features inside a 3D Gaussian scene. Our goal is not to train a new per-scene language classifier. Instead, we ask whether a scene representation can support two complementary interfaces from the same compact field: dense novel-view feature maps for rendered open-vocabulary localization, and registered primitive/point-level features for direct 3D selection or semantic querying. This feature-reconstruction framing differs from methods that attach only a final language code to a radiance field or optimize a single downstream score.

The technical challenge is that RADIO features are high-dimensional, dense, and expensive to store directly on every Gaussian. We therefore learn a compact Hybrid Gaussian Code Field, decode it through a Compact-to-Teacher Reconstruction module, and refine it in view space before comparison to the frozen RADIO teacher. We also use Frozen Geometry-Head Consistency as a geometry-aware regularizer: a frozen downstream head guides the feature field without turning the model into a task-specific predictor.

Our current paper package is conservative. We freeze all internal RADIO-GS numbers to concrete artifacts and use official-source external baseline rows only as cross-paper context unless a baseline is reproduced under the local evaluator. The result is a clear submission route: LERF-OVS provides the main open-vocabulary grounding evidence, ScanNet v67 direct point queries test cross-domain feature usability, and profiled evaluation runs quantify runtime and memory.

Contributions. This draft makes four claims:

- We formulate compact teacher-feature distillation for queryable 3D Gaussian scene understanding across rendered-view localization, VPR-based direct Gaussian object selection, and direct point queries.
- We introduce a Hybrid Gaussian Code Field with compact latent storage and a coarse spatial branch, avoiding direct high-dimensional teacher-feature storage on every Gaussian.

- We use Compact-to-Teacher Reconstruction, View-Space Feature Alignment, and Frozen Geometry-Head Consistency to recover teacher-compatible features while keeping teachers and downstream heads fixed.
- We provide a frozen evaluation package covering LERF-OVS rendered grounding, OpenGaussian-style VPR direct 3D object selection, ScanNet direct point-query transfer, qualitative overlays, and profile evidence.

2 Related Work

Language fields and open-vocabulary 3D understanding. LERF embeds language-aligned features in neural radiance fields and enables open-vocabulary querying in 3D scenes [5]. LangSplat and Language Embedded 3D Gaussians adapt this direction to Gaussian scene representations [13, 15]. OpenGaussian formalizes a stronger primitive-level protocol in which text selects 3D Gaussians that are then rendered only for mask evaluation [17]. Recent methods push this direction further: Dr. Splat registers language-aligned features directly to dominant Gaussians along pixel rays [6], CAGS introduces spatial context to reduce fragmented Gaussian representations [16], and InstanceGaussian and OpenSplat3D move toward instance-level Gaussian grouping and open-vocabulary instance segmentation [7, 12]. OpenInsGaussian further studies context-aware cross-view fusion for instance Gaussian segmentation [3], while very recent Ilov3Splat uses view-consistent feature fields and two-stage 3D clustering for instance-level open-vocabulary querying [10]. SuperGSeg clusters Gaussians into structured super-primitives for hierarchical understanding [9]. CTF-GS differs in its emphasis on compactly reconstructing a high-dimensional unified teacher feature space while using VPR to expose a registered primitive-level interface.

Gaussian scene representations. 3D Gaussian Splatting provides an explicit and efficient scene representation for radiance-field rendering [4]. CTF-GS keeps this geometry backbone and learns a feature field on top of it, separating geometric visibility from foundation-feature reconstruction.

Agglomerative vision foundation models. RADIO-style models distill multiple visual capabilities into a single dense vision backbone [14, 11]. CTF-GS uses C-RADIOv4-H as a frozen teacher and studies whether its dense features can be reconstructed from 3D scenes at novel views.

Foundation-model regularization in 3DGS. FMGS embeds 2D foundation-model features into Gaussian scenes and uses DINO-style spatial structure to improve feature consistency [18]. SAM-based Gaussian methods such as SAGA distill 2D segmentation priors into 3D Gaussian features for promptable object segmentation [1]. Our adaptor regularizers follow the same principle, but use the official C-RADIOv4-H adaptor heads so that the reconstructed feature remains a single RADIO-compatible representation at inference time.

3 Method

3.1 Problem Setup

Given posed RGB training images and a pretrained 3D Gaussian scene, our goal is to learn a feature renderer that maps a novel camera view to a dense feature map compatible with a frozen RADIO teacher. Let I_v be an image at view v and $T(\cdot)$ be the frozen RADIO encoder. The teacher target is:

$$F_v^\star = T(I_v) \in \mathbb{R}^{H \times W \times 1280}. \tag{1}$$

We learn a Gaussian feature field that renders $\hat{F}_v$ and minimizes feature reconstruction losses against $F_v^\star$.

3.2 Hybrid Gaussian Code Field

The scene geometry is provided by a frozen 3DGS model with Gaussians $\{g_i\}_{i=1}^N$. CTF-GS attaches compact learnable latent features to the Gaussians and combines two paths:

1. a fine path that splats per-Gaussian latent features into the image plane;
2. a coarse path that queries a spatial feature branch from reconstructed 3D positions.

The two paths are fused into a compact feature map Z_v. A Compact-to-Teacher Reconstruction module (implemented as the HCD codec in the codebase) maps this compact representation back to RADIO feature space:

$$\hat{F}_v = D_\theta(Z_v). \tag{2}$$

A View-Space Feature Aligner (implemented as the screen-space refiner in the codebase) then improves local alignment using rendered guidance signals such as visibility, alpha, and geometric depth.

3.3 Training Objective

The base feature loss combines cosine and reconstruction terms:

$$\mathcal{L}_{feat} = 1 - \frac{\langle \hat{F}_v, F_v^\star \rangle}{\|\hat{F}_v\|_2 \|F_v^\star\|_2} + \lambda_{rec} \|\hat{F}_v - F_v^\star\|_1. \tag{3}$$

For the FGC route, we add a frozen depth-head consistency loss. Let h_d be a frozen depth head and d_v^{geo} be the geometric depth rendered from the 3DGS backbone. We optimize:

$$\mathcal{L} = \mathcal{L}_{feat} + \lambda_{fgc}\mathcal{L}_{si}(h_d(\hat{F}_v), d_v^{geo}), \tag{4}$$

where $\mathcal{L}_{si}$ is a scale-invariant depth loss. The head is frozen; the feature field must adapt so that its rendered features remain useful to the head.

3.4 Frozen Adaptor Consistency

C-RADIOv4-H exposes frozen adaptor heads for `siglip2-g`, `dino_v3`, and `sam3`. The main open-vocabulary evaluator uses the text-aligned `siglip2-g` path, while training can additionally supervise the rendered feature through the other adaptors. For an adaptor A_a, we add a pointwise consistency term:

$$\mathcal{L}_a = 1 - \cos(A_a(\hat{F}_v), A_a(F_v^\star)). \tag{5}$$

For `dino_v3`, we also match the token self-similarity structure rather than only the local feature value. Given normalized adaptor tokens D and $D^\star$, we compute:

$$\mathcal{L}_{rel} = \|DD^\top - D^\star D^{\star\top}\|_2^2. \tag{6}$$

This transfers DINO-like boundary and correspondence structure into the rendered RADIO feature field, similar in spirit to DINO regularization in FMGS. We also evaluate a cross-view variant inspired by ProFuse's use of dense cross-view correspondence and context fusion [2]. For two training views p and q, we match teacher and rendered DINO token affinity:

$$\mathcal{L}_{cv} = \|D_p D_q^\top - D_p^\star D_q^{\star\top}\|_2^2. \tag{7}$$

Unlike ProFuse, this is an optimization-time regularizer on rendered feature maps, not a full pre-registration or context-proposal pipeline. To prevent cross-view smoothing from erasing text-grounding peaks, we add a label-free SigLIP text-heatmap distillation term over a fixed LERF text bank. The per-pixel query variant preserves the scene-category distribution:

$$\mathcal{L}_{txt}^{query} = \mathrm{KL}(\mathrm{softmax}(S^\star T^\top/\tau) \,\|\, \mathrm{softmax}(\hat{S}T^\top/\tau)). \tag{8}$$

We also evaluate a spatial variant that normalizes each query over image locations, directly targeting the LERF argmax localization failure mode:

$$\mathcal{L}_{txt}^{spatial} = \mathrm{KL}(\mathrm{softmax}_{xy}(s_q^\star/\tau) \,\|\, \mathrm{softmax}_{xy}(\hat{s}_q/\tau)). \tag{9}$$

For `sam3`, the current implementation uses a mask-free soft-region loss. Teacher SAM3 tokens define deterministic anchor prototypes; each pixel is softly assigned to anchors by teacher similarity, and the rendered feature is trained to match the teacher region prototypes:

$$\mathcal{L}_{reg} = \sum_k \|\mu_k(A_{\mathrm{sam3}}(\hat{F}_v)) - \mu_k(A_{\mathrm{sam3}}(F_v^\star))\|_2^2. \tag{10}$$

This approximates SAM-style region supervision when external SAM mask caches are not available, while keeping the inference-time representation unchanged.

3.5 Warm-Start Protocol

Directly applying frozen-head losses early in training can conflict with feature reconstruction. The mainline therefore uses a warm-start schedule: first train a feature-only field, then initialize the FGC model from that checkpoint and continue training with the frozen geometry-head loss active. This makes FGC a late geometry-aware refinement signal instead of an early competing objective.

3.6 Open-Vocabulary Grounding

At evaluation time, CTF-GS renders a feature map for each novel view. For a text query q, we compute a text embedding e_q and score each pixel by feature-text similarity with a scene-category softmax temperature τ. The predicted localization point is the heatmap maximum. LERF-OVS reports localization accuracy and mIoU over query heatmaps.

For direct 3D querying, the base Gaussian-center readout decodes pre-refiner compact features at Gaussian centers:

$$\hat{f}_i = D_\theta(c_i), \qquad s_i(q) = \cos(A_{\mathrm{SigLIP2}}(\hat{f}_i), e_q),$$

where A_{SigLIP2} is the frozen text-aligned summary head. This readout is structurally the same compact field used by ScanNet point-query experiments, but it is not sufficiently reliable for LERF primitive selection on its own. We therefore introduce *View-to-Primitive Registration* (VPR) for the LERF direct-selection protocol. VPR renders teacher-space features from posed views, applies VFA and the frozen SigLIP2 head, registers those text-aligned features back to visible Gaussian centers with depth/alpha checks, and then runs the same primitive-level text query on the registered Gaussian embeddings. A GT-free voxel-max context aggregation step propagates high primitive scores within local 3D neighborhoods to reduce fragmentation before selection. The LERF masks are used only for final query-select-render evaluation, not for registration, calibration, or scoring.

4 Experiments

4.1 Submission-Freeze Protocol

The conservative RADIO-GS numbers come from the 2026-05-02 submission-freeze report: `output/radio_gs/reports/submission_freeze_report.md`. The RADIO adaptor ablations were completed on 2026-05-03 under the same LERF sweep protocol. External baseline rows are now official-source values from the cited papers or supplements and are captioned as cross-paper context rather than reproduced same-evaluator results; the exact source rows and protocol caveats are recorded in `baseline_source_verification.md`.

Table 1: Frozen CTF-GS LERF-OVS rendered-feature grounding result. mIoU uses a GT-free peak-relative mask threshold of 0.60 selected by the global boundary-calibration sweep in Table 3; LocAcc is unchanged from the heatmap argmax.

Scene	LocAcc	mIoU	Temp.
Figurines	0.8214	0.4244	50
Ramen	0.9014	0.6201	40
Teatime	0.8983	0.5760	25
Waldo Kitchen	0.8636	0.4769	25
Macro	0.8712	0.5243	–

Table 2: LERF-OVS open-vocabulary grounding comparison. External rows are official-source LocAcc values converted to decimals; CTF-GS is the local rendered-feature result. Best per column is in **bold**.

Method	Figurines	Ramen	Teatime	Waldo Kitchen	Macro
LERF	0.795	0.625	**0.938**	0.815	0.793
LangSplat	0.804	0.732	0.881	**0.955**	0.843
LEGaussians	0.767	0.737	0.683	0.523	0.678
RADIO-GS	**0.821**	**0.901**	0.898	0.864	**0.871**

4.2 LERF-OVS Main Result

Table 1 is the current main internal result. Ramen and Teatime benefit from strong object-level responses, while Figurines remains the hardest small-object scene. The default threshold-0.50 readout gives 0.4941 macro mIoU; the global threshold-0.60 calibration tightens mask boundaries and raises the paper-facing macro mIoU to 0.5243 without changing localization accuracy. The adaptor-enhanced candidate in Table 12 is reported as an ablation/selector result rather than replacing the calibrated main row.

Table 2 provides the paper-facing external comparison. The LERF, LangSplat, and LEGaussians rows are traced to official paper or supplement tables and converted to decimal LocAcc when originally reported as percentages. Because these rows are not rerun under the RADIO-GS evaluator, we use them as context for the main claim rather than as a strict reproduced-SOTA leaderboard.

Table 3 isolates the rendered-mask boundary readout from feature quality. Raising the GT-free peak-relative mask threshold from 0.50 to 0.60 tightens the predicted region and improves macro mIoU on the paper-facing checkpoints from 0.494 to 0.524 without changing LocAcc. A threshold of 0.65 is marginally higher in macro mIoU but has lower weighted mIoU and visibly over-shrinks Figurines, so we use 0.60 as the safer global calibration for the main table and rendered qualitative masks. RGB boundary snap is more useful for direct 3D selection, where the rendered binary mask is only the final evaluation projection of already selected 3D primitives. An adaptive mean+std threshold diagnostic keeps LocAcc unchanged but lowers macro mIoU to 0.4939, so it is not promoted.

Table 3: GT-free rendered-mask boundary calibration on the paper-facing LERF-OVS checkpoints. All rows use the same checkpoints, text embeddings, temperatures, scoring, heatmap upsampling, and rendered heatmaps; only the peak-relative binary mask threshold changes. LocAcc is unchanged because it is measured from the heatmap argmax before binarization.

Readout	Fig.	Ramen	Tea.	Waldo	Macro
Threshold 0.50	**0.431**	0.586	0.549	0.411	0.494
Threshold 0.55	0.430	0.611	0.568	0.451	0.515
Threshold 0.60	0.424	**0.620**	0.576	0.477	0.524
Threshold 0.65	0.415	0.619	**0.578**	**0.490**	**0.525**
Threshold 0.70	0.395	0.596	0.556	0.477	0.506

Table 4: LERF-OVS direct 3D object selection under an OpenGaussian-style query-select-render protocol. External values are the OpenGaussian official paper numbers; CTF-GS uses rendered-view registered primitive features from 128 all-pose VPR views with GT-free voxel_max context aggregation and GT-free selection floor=0.005, selection cap=0.018 and a SigLIP2 text head. RGB snap is a GT-free rendered-mask boundary refinement with silhouette threshold 0.60 and is reported separately from the primitive-selection protocol.

Method	Protocol	Fig.	Ramen	Tea.	Waldo	Macro
OpenGaussian	mIoU	0.393	0.310	0.604	0.227	0.384
CTF-GS	meanstd2p5 mIoU	0.483	0.466	0.504	0.237	0.423
CTF-GS	+ RGB snap mIoU	0.548	0.471	0.562	0.241	0.455
OpenGaussian	Acc@0.25	0.554	0.422	0.763	0.318	0.514
CTF-GS	meanstd2p5 Acc@0.25	0.821	0.718	0.814	0.409	0.691
CTF-GS	+ RGB snap Acc@0.25	0.786	0.746	0.864	0.409	0.701

4.3 LERF Direct 3D Object Selection

Table 4 evaluates the primitive-level bridge to 2024–2025 open-vocabulary 3DGS work. We follow the OpenGaussian-style query-select-render protocol: text similarity is computed on 3D Gaussian primitives, selected primitives are rendered to official LERF-OVS annotated views, and metrics are mIoU and Acc@0.25. This protocol is not the same as the rendered-view heatmap evaluation in Table 1; the query is performed in 3D and rendering is used only to compare selected primitives with 2D masks.

The original Gaussian-center readout reached 0.080 macro mIoU and 0.093 macro Acc@0.25 under an earlier top-10% selector, but only 0.012/0.009 under the same fixed top-2% selector used by the VPR ablation rows. The registration readout closes most of this gap without using LERF masks for scoring. With GT-free voxel-max context aggregation, the conservative fixed top-2% selector reaches 0.385/0.643, while increasing the VPR view budget to 128 posed views and using the GT-free

Table 5: Published-context LERF-OVS direct 3D object-selection numbers. Values are reported as percentages from the cited papers or expert-traced source tables and are not same-evaluator local reruns. This table is only for positioning relative to recent primitive-/instance-level methods.

Method	Source type	mIoU	Acc@0.25
OpenGaussian [17]	official paper	38.36	51.43
Dr. Splat [6]	published context	43.29	64.30
CAGS [16]	published context	50.79	69.62
InstanceGaussian [7]	published context	45.30	58.44
OpenGaFF [8]	arXiv context	54.36	80.84
CTF-GS + VPR	local, SigLIP2, mean+2.5std+floor0.005+cap0.018	42.27	69.06
CTF-GS + VPR + RGB snap	local, optional GT-free rendered-mask boundary refinement	45.54	70.14

Table 6: VPR protocol card for LERF-OVS direct 3D object selection.

Item	Setting
Query target	3D Gaussian primitives; rendering is used only after selection for mask evaluation
Registration feature	Rendered RADIO-compatible features after VFA, projected by frozen SigLIP2 summary head
Registration views	Posed RGB/render views selected from all COLMAP poses; main row uses 128 evenly spaced views
Leakage audit	Train-view-only 24-view row is reported separately and remains above raw center readout
Visibility hygiene	Depth and alpha checks; no LERF masks or text labels are used during registration
Text scoring	Frozen SigLIP2 text/query embeddings with scene-wise softmax scoring
Selection	Fixed mean+2.5std selector with 0.5% floor and 1.8% cap; fixed-ratio sweep is diagnostic only
Context aggregation	GT-free voxel-max score aggregation, resolution 80, blend 0.50
Persistent storage	No extra trained state; optional registered primitive/voxel caches are reported separately
Metrics	OpenGaussian-style mIoU and Acc@0.25 after rendering selected primitives

mean+2.5std score-distribution selector with a 0.5% floor and 2% top-ratio cap reaches 0.419 macro mIoU and 0.690 macro Acc@0.25. Tightening the global cap to 1.8% improves the promoted row to 0.423/0.691, while 1.75% remains a near-tie diagnostic at 0.423/0.686 and 1.5% gives an accuracy-oriented diagnostic at 0.418/0.701. Applying the optional GT-free RGB snap only to the rendered evaluation mask improves the promoted 1.8% row from 0.423/0.691 to 0.455/0.701 by aligning the final projection with image boundaries. This is above the OpenGaussian official macro reference while using a different SigLIP2 text head and no locally rerun baseline reproduction; Waldo Kitchen also shows that explicit instance grouping can still be valuable on fragmented objects. We therefore use the result as evidence for registered primitive-level usability rather than as a universal primitive-level SOTA claim.

Table 5 gives the broader published context requested by the protocol audit. These rows are deliberately separated from the local result table: they mix papers, text heads, and implementation details, so they motivate positioning against recent primitive-/instance-aware methods rather than serving as a strict same-evaluator leaderboard.

Table 6 makes the direct-3D protocol auditable. The registration step uses 128 posed rendered views and visibility checks, but no LERF object masks or query labels. The paper-facing selector uses a fixed mean+2.5std multiplier on each query's primitive-score distribution; fixed-ratio sweeps, best-by-scene choices, and component-refinement variants are diagnostic.

Table 7 audits the implementation choices and disambiguates the two raw-center diagnostics: 0.080/0.093 is the earlier top-10% selector, whereas 0.012/0.009 is the same top-2% selector used by the VPR ablation rows.

Table 7: VPR diagnostics for LERF-OVS direct 3D object selection. Except for the explicitly marked historical raw-center row, rows use the same fixed top-2% primitive selector. "Reg." is the mean fraction of Gaussians receiving registered rendered-view features; unregistered primitives use the direct Gaussian-center fallback unless disabled by the ablation.

Variant	Selector / views	Reg.	mIoU	Acc@0.25
Raw Gaussian-center	top10%, previous	–	0.080	0.093
Raw Gaussian-center	top2%, VPR protocol	–	0.012	0.009
VPR, cosine	top2%, all poses 24	0.385	0.312	0.547
VPR, softmax	top2%, all poses 24	0.385	0.342	0.555
VPR + voxel	top2%, train views 24	0.369	0.349	0.595
VPR + voxel	top2%, all poses 8	0.240	0.277	0.487
VPR + voxel	top2%, all poses 48	0.448	0.380	0.653
VPR + voxel	top2%, all poses 96	0.485	0.385	0.643
VPR + voxel, lower blend	top2%, all poses 96	0.485	0.385	0.631
VPR + voxel, w/o VFA	top10%, all poses 24	0.385	0.070	0.099
VPR + voxel, w/o depth check	top2%, all poses 24	0.972	0.347	0.615

Table 8: Query-level audit for LERF direct 3D object selection. Confidence intervals are bootstrap intervals over query masks for the fixed meanstd2p5 selector.

Scene	Queries	mIoU	Acc@0.25	Zero pred.
Figurines	56	0.548	0.786	0.071
Ramen	71	0.471	0.746	0.042
Teatime	59	0.562	0.864	0.034
Waldo Kitchen	22	0.241	0.409	0.227

Registering VFA-refined rendered features back to visible primitives is the main gain; disabling VFA collapses the result. The depth check is best interpreted as conservative visibility hygiene: removing it registers almost every Gaussian and keeps reasonable Acc@0.25, but does not improve mIoU over the promoted 96-view VPR row. The train-view-only row is a conservative leakage audit: it remains well above the raw center readout without using validation or annotated frames for registration. We also tested Dr. Splat-inspired contribution-weighted registration using center-sampled alpha and alpha-depth weights. This lowered the fixed-protocol macro result to 0.298 and 0.297 mIoU, respectively, so we keep uniform VPR as the main row and leave true rasterization-contribution assignment as future work.

Table 8 audits the promoted direct-3D selector at query level. Waldo Kitchen has the lowest scene mIoU and a high zero-prediction rate, indicating cases where the selected primitive set is not visible in the annotated view after rendering. Ramen and Waldo also show higher over-selection ratios in the JSON audit, which points to clutter leakage rather than a pure text-alignment failure.

4.4 Rendered Features vs. Original RADIO Features

Table 9 supports the inference-time feature-memory claim: the rendered feature field is not merely a lossy copy of the teacher. It can sharpen the task-relevant scene

Table 9: Same-evaluator comparison between original RADIO features extracted from RGB frames and CTF-GS rendered features. Both use the same frozen SigLIP2 head, scene-category scoring, and GT-free threshold-0.60 mask readout.

Scene	RADIO RGB		CTF-GS	
	LocAcc	mIoU	LocAcc	mIoU
Figurines	0.7321	0.3816	0.8214	0.4244
Ramen	0.8873	0.5247	0.9014	0.6201
Teatime	0.8475	0.5192	0.8983	0.5760
Waldo Kitchen	0.7273	0.4280	0.8636	0.4769
Macro	0.7985	0.4634	0.8712	0.5243

Table 10: Controlled LERF-OVS component ablation on seed 7. The table reports rendered-feature LocAcc per scene plus macro LocAcc and mIoU.

Variant	Fig.	Ramen	Tea.	Waldo	LocAcc	mIoU
Full CTF-GS	0.768	0.901	0.898	0.864	0.858	0.485
w/o FGC warm-start	0.696	0.845	0.847	0.818	0.802	0.424
w/o VFA	0.768	0.859	0.915	0.818	0.840	0.480
w/o HGCF	0.768	0.873	0.898	0.818	0.839	0.507
w/o CTR	0.268	0.648	0.661	0.545	0.531	0.260

response after multiview reconstruction: under the same calibrated mask readout it improves both localization and macro mIoU over frame-wise RADIO RGB features, even though the original RADIO RGB feature remains the direct 2D teacher used during training.

4.5 Core Component Ablation

Table 10 isolates the main architectural choices on LERF-OVS with a controlled seed-7 protocol. Unlike Table 1, which reports the conservative current best per scene, this table keeps the same seed route for the full model and its ablations. The goal is to verify whether each component contributes under the same evaluator: HGCF tests compact feature storage, CTR tests the nonlinear compact-to-teacher bottleneck, VFA tests peak and boundary correction, and FGC warm-start tests the geometry-aware training stage.

The largest dependency is the CTR codec. Replacing CTR with a direct 1×1 projection drops macro LocAcc from 0.858 to 0.531 and macro mIoU from 0.485 to 0.260, with every scene degraded. FGC warm-start is the next strongest component, improving macro LocAcc by 0.056 and macro mIoU by 0.061 over the feature-only run. VFA gives a smaller but consistent localization gain (+0.018 macro LocAcc) with almost unchanged mIoU. The hybrid code field mainly affects peak stability: the explicit non-hybrid variant reaches higher macro mIoU (0.507) but lower macro LocAcc (0.839), improving region overlap on Figurines/Teatime while losing peak accuracy

Table 11: Storage footprint accounting for compact teacher-feature fields and optional VPR caches. Direct storage assumes one 1280-D fp16 teacher feature per Gaussian. Compact ckpt counts the stored feature-field model, CTR/HCD codec, and VFA/refiner tensors. VPR caches are inference-time artifacts, not persistent trained state.

Scene	#G	Direct MiB	Compact MiB	Saving	Opt. VPR cache MiB	Saving w/ cache
Figurines	168,791	412.1	237.0	1.74×	501.3	0.56×
Ramen	382,687	934.3	311.2	3.00×	1131.4	0.65×
Teatime	460,157	1123.4	338.1	3.32×	1360.4	0.66×
Waldo Kitchen	691,728	1688.8	418.5	4.04×	2050.3	0.68×

on Ramen/Waldo. This explains why some enhancement branches can improve mIoU without preserving LocAcc.

4.6 Storage Footprint

Table 11 closes the compactness claim with explicit footprint accounting. Direct storage assumes one 1280-D fp16 teacher feature per Gaussian and excludes ordinary RGB/geometry attributes. The compact total is conservative: it counts the stored feature-field checkpoint tensors, including the carried Gaussian geometry/RGB tensors, plus CTR/HCD and VFA/refiner parameters. Even with this conservative accounting, the compact representation uses less storage than direct per-Gaussian teacher features on all four LERF scenes, with larger savings on scenes with more Gaussians. VPR itself is an inference-time readout and does not add trained persistent parameters; if one materializes registered 1536-D SigLIP2 primitive embeddings and per-query voxel scores as a cache, that cache is reported separately and is larger than direct 1280-D fp16 feature storage. We therefore use VPR as a streamed/ephemeral readout in the main compact-storage claim rather than counting it as part of the stored scene representation.

4.7 Adaptor Consistency Ablation

Table 12 shows that adaptor supervision is useful but scene-dependent under the default threshold-0.50 readout. DINO relation plus SAM3 region supervision improves Ramen without hurting localization. Teatime also benefits from the relation/region variant, reaching 0.5592 mIoU at unchanged localization accuracy. For Figurines, DINO cross-view with spatial text-heatmap distillation recovers the baseline localization score while improving mIoU from 0.4308 to 0.4343. Waldo Kitchen remains on the baseline checkpoint. Using the promoted Figurines, Ramen, and Teatime adaptor/cross-view checkpoints while keeping baseline Waldo Kitchen gives a candidate adaptor-enhanced macro score of 0.8712 LocAcc and 0.4979 mIoU. Since the global boundary calibration in Table 3 supersedes this readout for the main LERF mask metric, adaptor consistency remains a method diagnostic rather than the promoted quantitative row.

Table 12: Full-sweep LERF-OVS adaptor ablations under the original threshold-0.50 readout. We only promote adaptor variants that preserve localization accuracy while improving mIoU; these diagnostics are kept separate from the calibrated Table 1 result.

Scene	Variant	LocAcc	mIoU
Figurines	baseline	0.8214	0.4308
Figurines	DINO rel. + SAM3 region	0.8036	0.4399
Figurines	DINO cv + spatial text	0.8214	0.4343
Ramen	baseline	0.9014	0.5862
Ramen	DINO rel. + SAM3 region	0.9014	0.5873
Teatime	baseline	0.8983	0.5486
Teatime	DINO rel. + SAM3 region	0.8983	0.5592
Waldo Kitchen	baseline	0.8636	0.4106
Waldo Kitchen	DINO + SAM3 pointwise	0.8636	0.4103
Macro	promoted selector	0.8712	0.4979

Table 14: Frozen adaptor-space downstream probes on LERF-OVS. "Teacher" uses original RADIO RGB features; "Rendered" uses CTF-GS rendered features. Prototype segmentation uses same-category support masks from other frames. Matching uses the first annotated frame as source.

Adaptor	Probe	Teacher		Rendered	
		LocAcc	mIoU	LocAcc	mIo
DINOv3	Prototype seg.	0.6543	0.0945	0.6277	0.09
DINOv3	Source-target match	0.5957	0.1032	0.5035	0.10
SAM3	Prototype seg.	0.8404	0.0757	0.6649	0.05
SAM3	Source-target match	0.7872	0.0953	0.7092	0.06

Table 13: ProFuse-inspired DINOv3 cross-view diagnostics on LERF-OVS under the original threshold-0.50 readout. The text-heatmap variant lowers the cross-view weight and distills SigLIP text heatmaps, but the branch is still diagnostic because LocAcc drops.

Scene / Variant	Temp.	LocAcc	mIoU
Figurines baseline	50	0.8214	0.4308
Figurines DINO cross-view	50	0.8036	0.4339
Figurines cross-view + text heatmap	50	0.8036	0.4381
Figurines cross-view + spatial heatmap	50	0.8214	0.4343
Waldo baseline	25	0.8636	0.4106
Waldo DINO cross-view	25	0.5909	0.3662
Waldo cross-view + text heatmap	40	0.6818	0.4563

Table 15: Promptable SAM3-adaptor and dense DINOv3-adaptor tasks on LERF-OVS. These probes use the same annotations as the grounding benchmark but replace text queries with adaptor-space prompts or source-target correspondences. SAM3 point/box prompts are prompt-constrained, mask propagation uses source-view support only, and no external SAM3 mask decoder is called. DINO mask propagation uses source-background contrast 0.5 plus GT-free robust readout: top-k foreground pooling, 2.0× propagated-area scaling, and peak-component cleanup.

Task	Teacher		Rendered	
	LocAcc/Hit	mIoU/Score	LocAcc/Hit	mIoU/Score
SAM3 point prompt	1.0000	0.3700	1.0000	0.4169
SAM3 box prompt	0.8702	0.6560	0.8221	0.6638
SAM3 mask propagation	0.7872	0.3583	0.6667	0.3756
DINOv3 dense matching	0.5723	0.8547	0.5393	0.9048
DINOv3 mask propagation + robust readout	0.7801	0.4806	0.7730	0.4456

The main failure mode is a localization/region tradeoff. The DINO relation and SAM3 region losses often smooth or broaden object-consistent heatmaps, which can raise thresholded overlap while moving the single argmax outside a small polygon. Because LERF-OVS LocAcc is an argmax-inside-mask metric, one feature-cell shift can hurt LocAcc even when the region response is visually cleaner. For submission, we therefore only promote adaptor checkpoints that preserve LocAcc; future training should add an explicit peak-preservation term or mask-aware small-object weighting before replacing the conservative main table.

Table 13 answers the cross-view question directly. DINO cross-view affinity and text-heatmap preservation can increase thresholded overlap, especially on Waldo at high temperature. The spatial heatmap variant recovers the Figurines localization score and becomes the only promoted cross-view LERF branch, but Waldo still loses the single-peak localization score. This suggests that cross-view affinity encourages coherent object regions while peak placement remains scene dependent; a full ProFuse-style pre-registration/context proposal module remains a future method extension rather than a paper-main result.

4.8 DINO/SAM Adaptor Downstream Probes

Table 14 is diagnostic rather than a claimed improvement: rendered features are competitive in DINOv3 mIoU but do not yet beat original RADIO features on aggregate, and unconstrained SAM3 prototype scoring exposes a larger teacher-rendered gap. Per-scene positives still exist, especially on Waldo Kitchen source-target matching, where rendered DINOv3 improves from 0.2500 to 0.5000 LocAcc and from 0.2708 to 0.2948 mIoU, and rendered SAM3 improves mIoU from 0.2965 to 0.3147 at unchanged LocAcc. These probes therefore support the feature-compatibility story while identifying adaptor-space preservation as the next optimization target. We therefore also report prompt-constrained task probes that better match the intended SAM3-adaptor and DINOv3-adaptor downstream use cases.

Table 15 separates two claims. First, rendered features improve prompt-constrained SAM3-adaptor mask mIoU over the frame-wise teacher for point prompts, box prompts, and mask propagation, while preserving perfect point-prompt localization. Second, source-background

Table 16: ScanNet v67 teacher-balanced direct point-query results. The contextual row uses kNN point readout with $k = 8$, 32 candidate Gaussians, and label-free scene-mean logit calibration with $\alpha = 0.5$.

Readout	Split	mIoU	mAcc
Gaussian-index	19 classes	0.3538	0.6076
Gaussian-index	15 classes	0.3573	0.6203
Gaussian-index	10 classes	0.4293	0.7051
Contextual kNN	19 classes	0.3637	0.6033
Contextual kNN	15 classes	0.3708	0.6224
Contextual kNN	10 classes	0.4512	0.7079

contrast makes DINOv3 mask propagation substantially stronger; adding robust foreground pooling, area-bounded propagation, and connected-component cleanup raises rendered DINO mask-propagation mIoU from the previous 0.3684 to 0.4456. This exceeds the previous fixed-readout teacher reference of 0.3921, but under the same robust readout the teacher also improves to 0.4806 mIoU and 0.7801 LocAcc. We therefore use this result as evidence that the DINO propagation gap is substantially narrowed, not as a same-protocol superiority claim. The higher rendered dense-matching similarity score suggests that the feature field can become over-smooth and select visually similar nearby regions. This motivates the cross-view DINO and text-heatmap protected training variants evaluated next. We also evaluate a mutual-correspondence plus homography-RANSAC diagnostic for qualitative DINO matching; it removes many outlier matches and raises rendered dense-match similarity to 0.9277, but it does not improve the same robust DINO mask-propagation mIoU beyond the teacher, so it remains a visualization and diagnostic filter rather than a main quantitative claim.

4.9 ScanNet Direct Point-Query Transfer

Table 16 evaluates cross-domain feature usability on 10 ScanNet scenes. The conservative block uses `gaussian_index` queries, `label_point` positions, and `label_index` opacity filtering. This is not presented as a full ScanNet semantic segmentation leaderboard result; it is a fair direct probe of the learned feature field. We also evaluate a label-free contextual point readout that queries each ScanNet vertex from nearby Gaussians instead of the row-aligned Gaussian index. This kNN readout improves the three split mIoUs to 0.3637, 0.3708, and 0.4512 while keeping mAcc comparable. Increasing the calibration scale to 0.75 further raises the 10-class mIoU to 0.4534 but lowers the 19-/15-class and mAcc averages, so we use $\alpha = 0.5$ as the balanced setting. ScanNet alias prompts remain mixed and are kept as appendix evidence.

Table 17 upgrades the earlier two-scene diagnostic to a full 10-scene ablation. DINO cross-view affinity improves the macro 19-class and 15-class probes more clearly than the 10-class probe, with positive macro mIoU and mAcc on all splits. The effect remains scene-dependent: the DINO branch improves 6/10, 7/10, and 5/10 scenes on the 19-, 15-, and 10-class mIoU splits, respectively. We therefore report it as a compatibility/FAC ablation rather than replacing the conservative main ScanNet row in Table 16.

Table 17: Full 10-scene ScanNet DINOv3 cross-view ablation. The branch is warm-started from the v67 teacher-balanced checkpoint and uses conservative DINO cross-view affinity weight $\lambda = 0.001$ with batch size 2.

Split	Base mIoU	DINO mIoU	Δ	Base mAcc	DINO mAcc	Δ
19 classes	0.3538	0.3640	+0.0102	0.6076	0.6205	+0.0128
15 classes	0.3573	0.3662	+0.0089	0.6203	0.6313	+0.0110
10 classes	0.4293	0.4308	+0.0014	0.7051	0.7071	+0.0020

Table 18: Submission efficiency and cost evidence. Runtime rows use explicit freeze-profile telemetry; storage rows compare direct per-Gaussian 1280-D fp16 teacher features against the stored compact feature-field footprint.

Evidence	Scope	Cost	Peak VRAM
LERF eval	4 scenes	124.770 s / 31.193 s-scene	2076 MiB
ScanNet eval	10 scenes	150.903 s / 15.090 s-scene	1666 MiB
Storage	4 scenes	1.74×–4.04× saving	–

4.10 Efficiency and Cost

Table 18 separates profiled evaluation cost from storage footprint accounting. The four LERF overlay evaluations take 124.770 seconds in total and peak at 2076 MiB, while the full 10-scene ScanNet v67 direct-query evaluation takes 150.903 seconds and peaks at 1666 MiB. Training throughput is kept as supplementary log-derived evidence because training wall time and explicit GPU telemetry are different measurement types.

4.11 Qualitative Results

The current qualitative shortlist is regenerated from the calibrated rendered mask readout used in Table 1. The recommended main figure uses one frozen rendered grid from each LERF scene: Figurines frame 00152, Ramen frame 00024, Teatime frame 00140, and Waldo Kitchen frame 00089.

4.12 Failure Analysis

Table 19 turns the qualitative small-object observation into a paper-facing diagnostic. The lowest-ranked categories are mostly tiny or visually ambiguous objects, such as Figurines bags, character toys, and isolated kitchen tools. The failure pattern is not only low region overlap: some categories show nonzero mIoU but zero or partial LocAcc, which means the rendered heatmap covers a

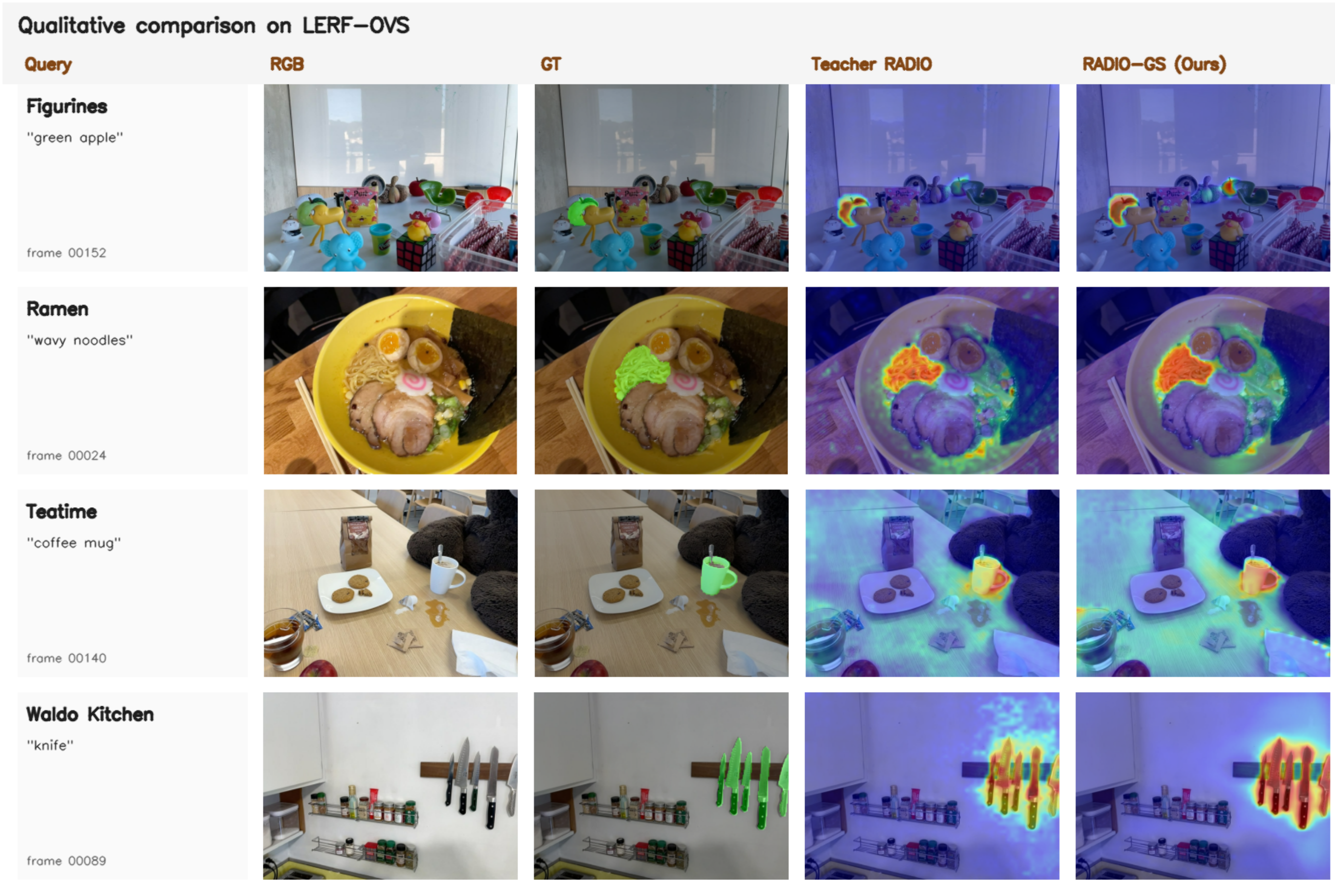

Figure 1: Qualitative open-vocabulary grounding examples from the calibrated LERF overlay runs. Each row shows the RGB image, ground-truth mask overlay, frozen C-RADIOv4-H teacher heatmap, and rendered CTF-GS heatmap for a representative query. Rendered masks use the same GT-free threshold-0.60 boundary calibration as Table 1.

Table 19: Representative fragile LERF-OVS categories from the frozen rendered-feature evaluation. Low LocAcc with nonzero mIoU highlights the peak-vs-region tradeoff.

Scene	Category	LocAcc	mIoU
Figurines	bag	0.00	0.00
Figurines	miffy	0.00	0.01
Teatime	dall-e brand	0.00	0.01
Figurines	jake	0.00	0.02
Waldo Kitchen	ketchup	0.00	0.03
Waldo Kitchen	spatula	0.00	0.03
Ramen	corn	0.00	0.10
Figurines	red toy chair	0.00	0.13

plausible object region while the single maximum lands outside the annotated polygon. This is why the paper treats mIoU improvements from adaptor or cross-view losses conservatively unless localization accuracy is preserved.

5 Discussion

Why feature reconstruction matters. The LERF-OVS results show that rendered RADIO-like features retain enough semantic structure for open-vocabulary localization. The ScanNet results suggest that the same representation can be probed outside the LERF object-query setting. The LERF direct-selection experiment further clarifies the boundary of this claim: the compact field supports direct primitive querying once rendered features are registered back to Gaussians, but the current voxel context is still weaker than explicit instance- or super-Gaussian grouping on fragmented scenes.

External-baseline scope. The LERF, LangSplat, and LEGaussians comparison rows are now traced to official paper or supplementary tables. They still come from cross-paper protocols, so the paper treats them as source-anchored context rows. A strict SOTA claim would require rerunning those baselines under the same evaluator and annotations used by RADIO-GS.

6 Limitations

CTF-GS depends on a pretrained Gaussian geometry backbone and does not claim to improve RGB reconstruction. Small objects remain challenging because the teacher feature resolution and rendered feature alignment limit heatmap sharpness. The current ScanNet protocol is a direct point-query feature probe, not a replacement for a full standardized segmentation benchmark. The registered LERF direct 3D object-selection readout substantially improves primitive querying and slightly exceeds the OpenGaussian official macro reference under the reported OpenGaussian-style context, but it remains weaker on Waldo Kitchen and is not a locally rerun same-evaluator comparison. The paper should therefore avoid claiming universal primitive-level SOTA. Finally, the external comparison is official-source but not reproduced under one evaluator, so strong SOTA-style claims require additional baseline reruns. The storage table reports checkpoint footprint rather than a deployed runtime memory allocator, so future packaging should separate feature-field parameters, rendering buffers, and optional downstream heads.

7 Conclusion

CTF-GS turns a 3D Gaussian scene into a reusable foundation-feature memory. By reconstructing frozen RADIO features rather than training a scene-specific classifier, it supports open-vocabulary grounding and cross-domain feature queries from novel views. The current freeze package provides a coherent submission path: strong LERF-OVS internal results, fair ScanNet direct-query evidence, explicit storage and runtime profiles, failure analysis, and source-anchored external comparison caveats.

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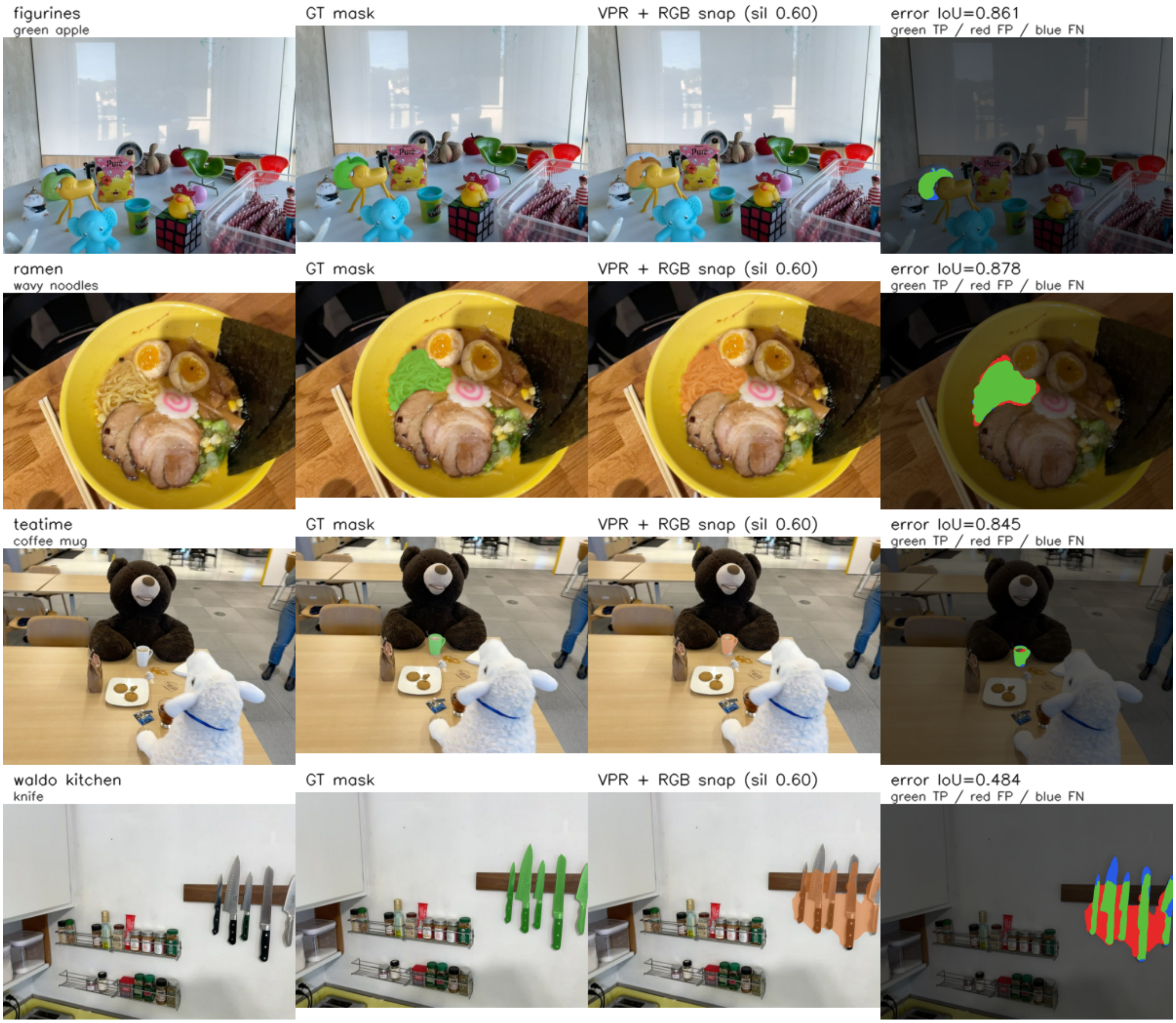

Figure 2: Qualitative LERF direct 3D object selection with VPR. Each row fixes a scene, annotated frame, and query, then shows RGB, ground-truth mask, the rendered mask from the promoted mean+2.5std VPR selector with a 0.5% floor and 1.8% cap after GT-free RGB snap with silhouette threshold 0.60, and an error map. Green is true positive, red is false positive, and blue is false negative. The Waldo Kitchen row illustrates the remaining fragmentation and clutter failure mode.

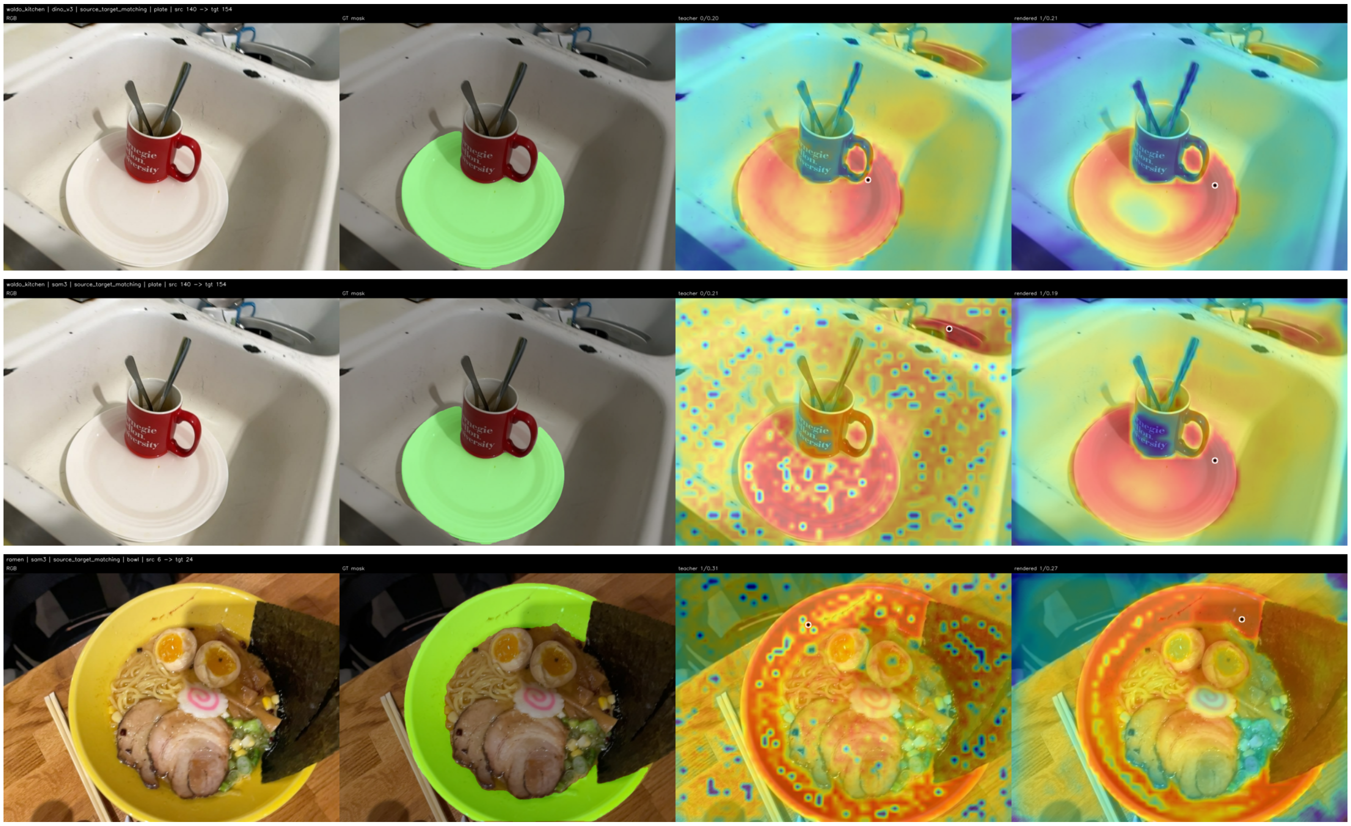

Figure 3: Qualitative DINOv3/SAM3 adaptor-space source-target matching. Each row shows RGB, GT mask, original RADIO teacher heatmap, and CTF-GS rendered heatmap. The Waldo Kitchen rows show cases where rendered features move the peak onto the target object; the Ramen row illustrates that the aggregate SAM3 gap remains visible even when localization is correct.

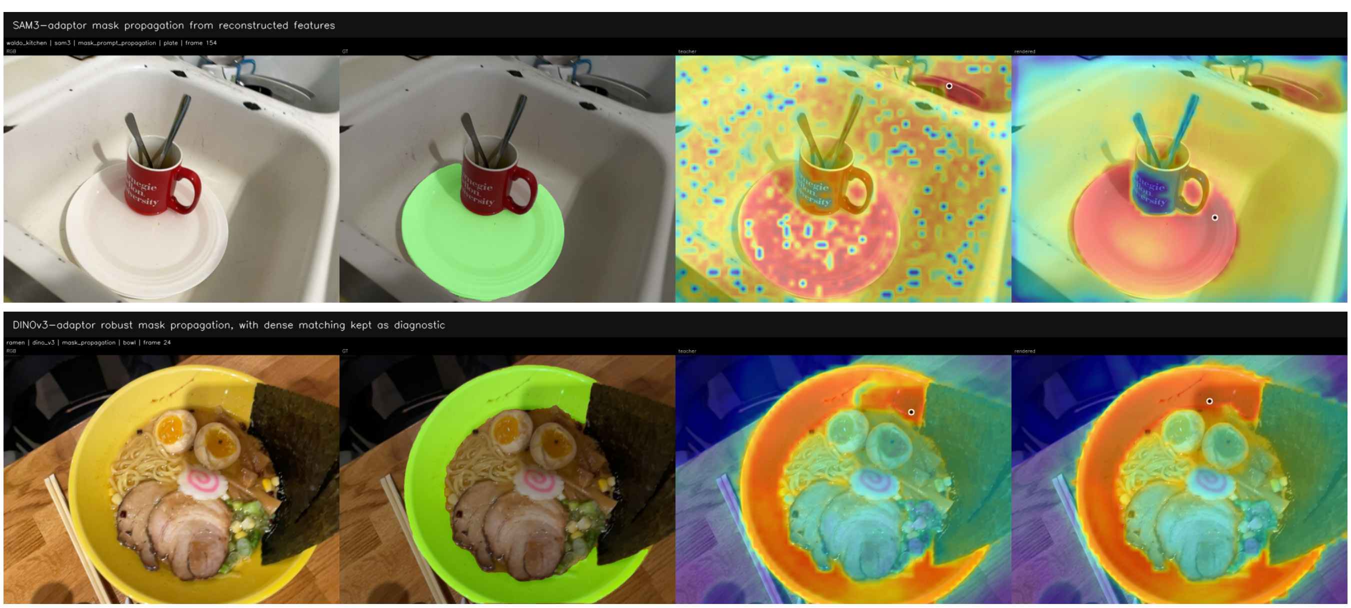

Figure 4: Qualitative promptable downstream probes from reconstructed features. Top: SAM3 mask-prompt propagation from a source annotation to a target view. Bottom: DINOv3 dense matching visualized with teacher and rendered correspondences. The paper-facing diagnostic can optionally apply mutual matching plus homography RANSAC to suppress geometric outliers in this bottom row.